



Data Article

SkinDisNet: A comprehensive dataset of clinical images and metadata for skin disease detection and classification



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ABSTRACT

Skin diseases are considered among the most important health problems worldwide, affecting millions of people. Early diagnosis can prevent complications and therefore improve the outcomes for patients. However, accurate diagnosis can be challenging due to the visual similarity of different conditions, variability in symptoms, and the shortage of region-specific annotated datasets for developing robust machine learning models. To bridge the gap, this paper presents a new dataset that aims to further the field of skin disease detection through machine learning and computer vision techniques. There are 1710 clinical images of skin diseases, captured using smartphones cameras from hospitals in Bangladesh. The images are organized into six classes of skin diseases—Atopic Dermatitis, Contact Dermatitis, Eczema, Scabies, Seborrheic Dermatitis, and Tinea Corporis. Each image is uniquely identified and stored in categorized folders, with all entries reviewed to ensure accurate classification. Moreover, this paper includes metadata that has been curated meticulously to provide essential clinical and demo-

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graphic details, enabling precise analysis and reproducible research. Our dataset is intended to facilitate research and advancement in skin disease detection and classification using machine learning and computer vision techniques.

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Specifications Table

Subject	Computer Sciences
Specific subject area	Image Processing, Medical Image Analysis, Skin Disease Classification, Computer Vision
Type of data	Images (Preprocessed and Augmented) and metadata
Data collection	Clinical images of six categories of skin diseases were collected from Rangpur Medical College, Rangpur, and Shahid Syed Nazrul Islam Medical College, Kishoreganj, Bangladesh, between April 2023 and June 2024. Images were captured using smartphone cameras in natural light during patient consultations with dermatologists. The inclusion criteria required clear signs of skin disease, while ambiguous or obscured images were excluded. The metadata of the corresponding images are gathered according to the given information during patient's consultation.
Data source location	Institution: Rangpur Medical College City: Rangpur Country: Bangladesh Latitude: 25°46'00.2"N, Longitude: 89°14'01.9"E Institution: Shahid Syed Nazrul Islam Medical College City: Kishoreganj Country: Bangladesh Latitude: 24°24'56.1"N, Longitude: 90°47'21.5"E
Data accessibility	Repository name: Mendeley Data [1] Data identification number: 10.17632/yj3md44hxg.1 Direct URL to data: https://data.mendeley.com/datasets/yj3md44hxg/ Instructions for accessing these data: Under the license of CC BY-NC 4.0, the dataset is publicly available on Mendeley Data. Anyone can access and download the dataset through the URL to data for academic and research purposes only.
Related research article	None

1. Value of the Data

- This dataset is a useful resource to develop accurate and efficient methods for the detection and classification of skin diseases through machine learning and computer vision techniques.
- Researchers can utilize this dataset as training data to develop Computer-Aided Diagnosis (CAD) systems to deal with several skin diseases.
- This dataset can also be useful for education purposes. It can be used to train medical interns as well as students to identify different types of skin diseases.

2. Background

Skin diseases are classified as disorders that affect the skin, the largest organ in the body [2]. This is a common disease which can affect people of all ages and ethnicities. The World Health Organization (WHO) reported that in 2020, Bangladesh experienced 1131 fatalities attributed to skin diseases, accounting for 0.16 % of all deaths [3]. Early detection and treatment of skin diseases can prevent complications, reduce discomfort, and enhance an individual's overall quality of life [4]. Skin diseases vary in different regions due to different environmental factors and cli-

mate change, pollution and socioeconomic conditions, which influence the prevalence, type, and severity of these conditions [5]. In South Asian developing countries like Bangladesh, India and Nepal, there are a few datasets available for detecting skin disease, which hinders the development of accurate diagnostic tools. This dataset aims to bridge that gap by providing a resource to aid in the early detection and study of skin diseases in these regions.

3. Data Description

This article presents a clinical image dataset of skin diseases collected between April 2023 and June 2024, from Rangpur Medical College in Rangpur and Shahid Syed Nazrul Islam Medical College in Kishoreganj, two renowned medical colleges in Bangladesh. The dataset is organized into two main folders: Preprocessed and Augmented. Each folder is further categorized by the six skin disease classes: Atopic Dermatitis (AD), Contact Dermatitis (CD), Eczema (EC), Scabies (SC), Seborrheic Dermatitis (SD), and Tinea Corporis (TC). The comprehensive overview of the classes is presented in [Table 1](#). Each folder contains the images labelled with the disease's short name followed by a unique identifier such as AD (1).jpg, CD (5).jpg). Each image is labelled based on expert dermatologists' diagnoses, ensuring the dataset's accuracy and consistency are maintained. The images are accompanied by meticulously curated metadata, encompassing key clinical and demographic details, including patient age, sex, lesion location, and diagnosis. The augmented images labelled with an additional ID to distinguish between the different augmentation variations. The augmentation identifiers are as follows:

- 01: Rotation
- 02: Flipping (Horizontal & Vertical)
- 03: Scaling (Zoom)
- 04: Translation (Shift)
- 05: Brightness Adjustment
- 06: Contrast Adjustment
- 07: Shearing (on X & Y axes)

There are a total of 1710 preprocessed images and 11,970 augmented images of a total number of 416 patients in this dataset. The number of preprocessed and augmented images of each disease class is listed in [Table 2](#).

The primary goal of developing this skin disease dataset is to advance the study of automated detection and classification of dermatological conditions. This dataset facilitates the development of machine learning models, particularly in the field of image analysis and deep learning techniques.

4. Experimental Design, Materials and Methods

4.1. Data collection

The dataset consists of clinical images of six different categories of skin disease. Images were collected manually by the researchers from the dermatology outdoor department of hospitals during patients' consultation with dermatology specialists. They were captured using the default cameras of three different smartphones in natural lighting conditions without any prior set up. [Table 3](#) highlights the specifications of the devices employed for data collection. We also obtained patient consents to include their images in this study.

4.2. Preprocessing

The collected images exhibited a range of sizes and shapes. Therefore, a series of preprocessing was performed to the dataset for enhancing the quality and consistency of the im-

Table 1
Brief Overview of the Dataset.

Name of Disease	Description	Visualization
Atopic Dermatitis	Atopic dermatitis is a usual chronic inflammatory skin disorder which is notably itchy and characterized by dry, irritated skin. The condition is associated with skin barrier abnormalities, exposure to environmental as well as viral agents, and immune system dysregulation, leading to ongoing inflammation and severe itching [6]. Atopic dermatitis commonly appears in various body areas depending on age. In infants, it typically appears on the face and outer limbs, sparing the diaper area. From ages 1–2 years onwards, lesions are often found in skin folds. Young people and adults frequently have lichenified and excoriated plaques in flexural regions, including the wrists, ankles, and eyelids [7].	
Contact Dermatitis	Contact dermatitis refers to a widespread inflammatory dermatological condition resulting from recurrent exposure to allergies or irritants, which can lead to either allergic contact dermatitis (ACD) or irritant contact dermatitis (ICD). It can significantly impact the quality of life and work productivity, necessitating a better understanding for improved diagnosis and personalized treatment [8]. Contact dermatitis typically impacts particular regions of the body as a result of repeated exposure to irritants or allergens. These commonly affected areas include hands, face and neck, arms, trunk and buttocks, ears, waistline [9]. The specific locations depend on the type of exposure, such as chemicals, cosmetics, jewelry, or clothing materials [10].	
Eczema	Eczema is an inflammatory, non-infectious reaction in the epidermal and papillary layers of the skin, triggered by various external and/or internal factors, along with a heightened individual susceptibility. This is a highly prevalent form of inflammatory skin condition, impacting the functional unit of the epidermis along with the underlying dermal papillary layer, and is characterized by intolerance reactions in these areas [11]. Eczema often impacts visible areas such as the face, neck, eyelids, and around the lips, causing heightened anxiety and widespread itching in patients [12].	
Scabies	Scabies is a skin infestation in humans that occurs when the parasitic mite <i>Sarcoptes scabiei</i> var. <i>hominis</i> burrows into the epidermis [13]. It impacts millions of individuals around the world, particularly in areas with insufficient hygiene and densely populated living environments [14]. The disease primarily spreads through close physical contact and commonly affects areas such as the spaces between the fingers, wrists, armpits, abdomen, buttocks, and genital regions [15].	

(continued on next page)

Table 1 (continued)

Name of Disease	Description	Visualization
Seborrheic Dermatitis	Seborrheic dermatitis is a typical long-term inflammatory dermatological condition that primarily impacts young adults in regions abundant in sebaceous glands (scalp, face, and trunk). It presents with a range of manifestations, from mild patches to diffuse scaling on the scalp in adults and as yellowish, scaly patches on the scalp in infants (referred to as 'cradle cap') [16]. Seborrheic dermatitis usually impacts areas of the skin that produce a significant amount of sebum, such as the scalp, nasolabial folds, glabella, eyebrows, beard, ears, behind the ears, sternum, and various skin folds [17].	
Tinea Corporis	Tinea corporis, often referred to as "ringworm", is a surface dermatophyte infestation affecting the skin, with the exception of the hands, feet, scalp, bearded areas, face, groin, and nails. Its appearance is clearly defined, sharply delimited, round-shaped, moderately erythematous, scaly patch with a raised leading edge, and moderate itching is frequently observed [18]. Tinea corporis mainly targets the smooth, hairless skin on the trunk, arms, and legs, avoiding the palmoplantar skin and skin folds. This condition generally impacts the smooth skin of the trunk and limbs, while leaving the palms, soles, and skin folds unaffected [19].	

Table 2
Distribution of Dataset by Disease.

Name of Disease	Number of Original Images	Number of Augmented Images
Atopic Dermatitis (AD)	70	490
Contact Dermatitis (CD)	477	3339
Eczema (EC)	466	3262
Scabies (SC)	343	2401
Seborrheic Dermatitis (SD)	79	553
Tinea Corporis (TC)	275	1925

Table 3
Specifications of the devices utilized for collection of data.

Device Name	Camera Specification	Image Dimensions
Samsung Galaxy A50	25 megapixels	4032 × 3024
iPhone 14 Pro Max	48 megapixels	4032 × 3024
Realme C25s	48 megapixels	4208 × 3120

ages. At first, each diagnosed image was carefully labelled according to its corresponding skin disease and organized into a specific folder labelled by the disease name. Afterwards, the images were cropped into a square shape to focus on the disease area by removing unnecessary background elements. Then unwanted objects and backgrounds were removed to enhance the further quality. This involved eliminating the distractions like clothes, shoes, etc. Finally, after

cleaning the images, all the images are resized to the uniform size of 512×512 pixels for consistency.

4.3. Augmentation

Data augmentation techniques were applied to the dataset to enhance the model's ability to generalize across variations in image acquisition. This included both geometric and photometric transformations. The geometric transformations were implemented using affine matrices to simulate spatial variations [20]. Rotation up to $\pm 50^\circ$ was applied using the 2D rotation matrix shown in Eq. (1):

$$R(\theta) = \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix} \quad (1)$$

Translation, involving horizontal and vertical shifts of up to 20 % of the image dimensions, was modeled as the following Eq. (2):

$$T(w, h) = \begin{bmatrix} 1 & 0 & w \cdot W \\ 0 & 1 & h \cdot H \end{bmatrix} \quad (2)$$

where $w, h \in [-0.2, 0.2]$, and W and H represent the image width and height, respectively.

Zooming transformations were applied using a uniform scaling matrix in Eq. (3):

$$Z(z) = \begin{bmatrix} z & 0 \\ 0 & z \end{bmatrix} \quad (3)$$

where $z \in [0.8, 1.2]$ represents the scaling factor applied uniformly along both the horizontal and vertical axes. Shearing with a shear factor $s = 0.2$, defined using an affine transformation in Eq. (4):

$$S(s) = \begin{bmatrix} 1 & s & -s \cdot \frac{W}{2} \\ s & 1 & -s \cdot \frac{H}{2} \end{bmatrix} \quad (4)$$

Flipping operations were modelled as reflection matrices, with horizontal flipping and vertical flipping defined using Eq. (5) and Eq. (6) respectively:

$$H = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \quad (5)$$

$$V = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \quad (6)$$

Photometric transformations were applied to simulate variations in lighting and contrast conditions. Brightness adjustment and contrast enhancement were performed using the mathematical equations Eq. (7) and Eq. (8) respectively.

$$I' = b \cdot I \quad (7)$$

where I and I' denote the original and adjusted pixel intensities, respectively, and the brightness factor b was randomly sampled from the range $[0.7, 1.3]$.

$$I' = \alpha \cdot I + \beta \quad (8)$$

where $\alpha = 1.2$, $\beta = 0$, followed by clipping to maintain pixel values within a valid range.

These augmentations created multiple variations of each image, hence increasing the dataset and giving it more robustness. Such strategies have proven effective in deep learning pipelines for image classification tasks [21].

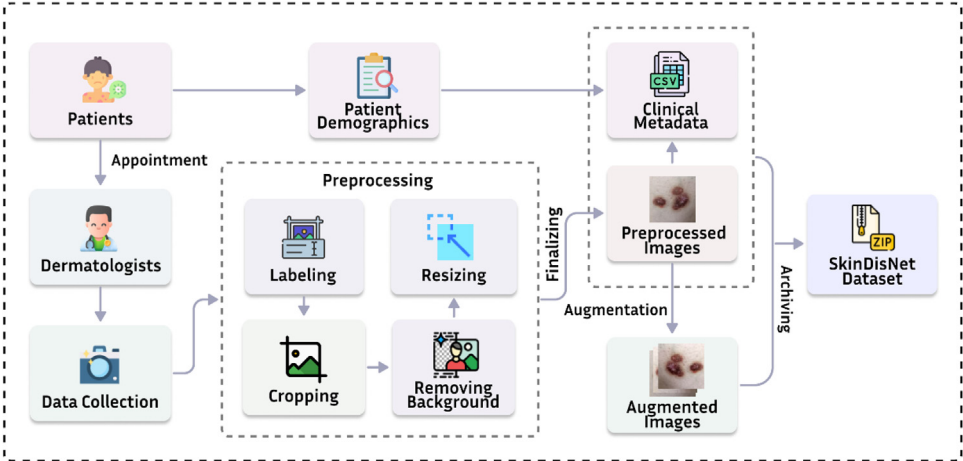


Fig. 1. Pipeline for Skin Disease Clinical Dataset Creation. The figure illustrates the process of creating a skin disease clinical image dataset, starting from data collection, followed by preprocessing (labelling, cropping, background removal, resizing), augmentation for diversity, and final archiving into a compressed dataset with metadata.

Table 4
Performance metrics of the three deep learning models.

Model Name	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
efficientnet_b0	86.93	82.52	84.71	83.2
resnet50	86.36	85.06	86.67	85.49
vgg16	86.93	88.12	86.76	87.34

5. Dataset construction

The dataset is organized into two main folders: Preprocessed Images and Augmented Images, each containing six skin disease classes. The preprocessed images folder contains clinically validated and meticulously labelled images and the augmented images folder features enhanced variations generated through augmentation.

Fig. 1 illustrates the pipeline of dataset creation, highlighting each step in the process.

5.1. Validation of deep learning models

We recommend utilizing a conventional deep learning model to effectively train the dataset, with the goal of achieving cutting-edge outcomes. The validation of a deep learning model involves evaluating its performance on a dataset through processes such as data splitting, model training, and performance analysis on an entirely different testing set. The dataset contains both preprocessed and augmented images which is further split into three subsets: training, validation and testing. After training on the training set, the model's hyperparameters and intermediate performance are tuned using the validation set, and the final performance of model is assessed using the testing set. The process of validating the deep learning model using this dataset is depicted in Fig. 2.

A simple classification experiment was conducted to validate the data quality and assess the suitability of the dataset for classification tasks. Therefore, we employed three baseline deep learning models—EfficientNetB0, ResNet50, and VGG16. The strong testing performance of these models underscores the efficacy of our dataset in facilitating precise classification. This evaluation's findings are presented in Table 4.

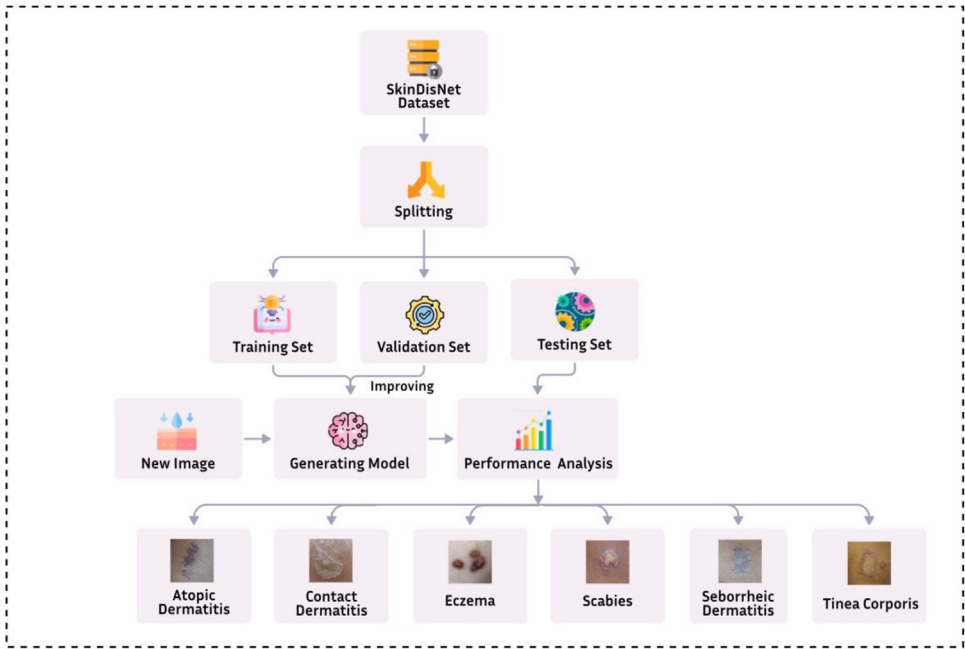


Fig. 2. Working process for classification of skin diseases.

Limitations

- The dataset was obtained from only two institutions in Bangladesh. Skin disorders exhibit regional variability influenced by environmental factors, climate change, pollution, and socioeconomic conditions, indicating that this dataset may not cover all types of skin diseases.
- Participation in this study was entirely voluntary, and several individuals may have decided not to provide images because of privacy concerns, personal reasons, or uncertainty about data utilization. If a substantial number of individuals declined to participate, this could result in a smaller dataset and possible selection bias.
- Individuals with mild or severe skin conditions may have exhibited varying likelihoods of participation, resulting in an imbalance in data representation. This self-selection bias may affect the dataset's ability to accurately represent all aspects of skin disease severity.
- The data were obtained using various smartphone cameras and collected at different times and days. This may result in variations in lighting, colour accuracy, and image clarity. Despite attempts to maintain consistency, these factors may lead to slight differences among images.

Ethics Statement

The authors ensured adherence to the ethical guidelines outlined in the Declaration of Helsinki, established by the World Medical Association, as it involved human subjects. It is clarified that patients' data were anonymized to protect their privacy, and no personal or identifiable information was contained in the dataset. The formal approval for data collection was obtained from the Principal of Syed Nazrul Islam Medical College, Kishoreganj, and the administration of Rangpur Medical College, Rangpur, before the study commenced and ethical approval has also been taken from affiliated department of corresponding author. However, the research's objective was disclosed to all participants, who subsequently granted consent to it. They were also

informed of the significance of the research prior to the beginning of data collection. The images were solely used for research purposes and will contribute to the development of the diagnosis and treatment of skin diseases.

Credit Author Statement

Marjia Sultana: Conceptualization, Data Curation, Supervision, Project administration, Funding acquisition, Investigation, Writing – Original draft preparation; **Rakib Hossain Sajib:** Methodology, Software, Data Curation, Formal Analysis, Investigation, Visualization, Writing – Original draft preparation; **Iffat Ara Badhan:** Writing – Reviewing and Editing; **Mizu Ahmed:** Investigation, Data Validation, Data Curation; **Moumita Yasmin Bushra:** Data Validation, Data Curation.

Data Availability

SkinDisNet: A Multi-Class Clinical Images and Metadata for Skin Disease (Original data) (Mendeley Data).

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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